Chapter 1. Set Theory and Logic

Sets and functions, relations, countability, recursive definition, the axiom of choice, well-ordering, Zorn's lemma

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Bracketed numbers after a result refer to Munkres.

1.1 Fundamental concepts

Motivation. Topology uses sets and functions heavily; this chapter fixes the language and the facts (countability, choice, well-ordering) that later proofs rely on.

We use sets, membership, inclusion $A \subset B$, union, intersection, difference $A - B$, the empty set, and the cartesian product $A \times B$ of ordered pairs. For a collection $\mathcal{A}$ of sets, $\bigcup_{A\in\mathcal{A}} A$ and $\bigcap_{A\in\mathcal{A}} A$ are defined as usual (the intersection of the empty collection is left undefined). De Morgan's laws hold:

$$A - \bigcup_\alpha B_\alpha = \bigcap_\alpha (A - B_\alpha), \qquad A - \bigcap_\alpha B_\alpha = \bigcup_\alpha (A - B_\alpha),$$

and $A \cap \bigcup B_\alpha = \bigcup (A \cap B_\alpha)$, $A \cup \bigcap B_\alpha = \bigcap (A \cup B_\alpha)$. In logic, "$P \Rightarrow Q$" is equivalent to its contrapositive "not $Q \Rightarrow$ not P" but not to its converse; the negation of "for every x, $P(x)$" is "for some x, not $P(x)$".

1.2 Functions

A **function** $f : A \to B$ is a rule (formally a subset of $A\times B$ in which each a appears as first coordinate of exactly one pair). For $A_0 \subset A$, $f(A_0)$ is the image; for $B_0 \subset B$, $f^{-1}(B_0) = \{a : f(a) \in B_0\}$. f is **injective** if $f(a) = f(a')$ implies $a = a'$, **surjective** if $f(A) = B$, **bijective** if both; a bijection has an inverse $f^{-1} : B \to A$. Composites of injections (surjections) are injections (surjections). Preimages preserve unions, intersections and differences; images preserve unions and inclusions only, and $A_0 \subset f^{-1}(f(A_0))$, $f(f^{-1}(B_0)) \subset B_0$, with equality if f is injective (resp. surjective).

Lemma 1.1 (2.1)

Let $f : A \to B$. If there are $g, h : B \to A$ with $g(f(a)) = a$ for all $a \in A$ and $f(h(b)) = b$ for all $b \in B$, then f is bijective and $g = h = f^{-1}$.

Proof. $f(a) = f(a')$ gives $a = g(f(a)) = g(f(a')) = a'$, so f is injective; $b = f(h(b))$ shows it is surjective. Then $g = g \circ (f \circ h) = (g \circ f) \circ h = h$, and both equal f^{-1}. ∎

1.3 Relations

Motivation. Equivalence relations will give quotient spaces and order relations the order topology, both introduced here set-theoretically.

A **relation** on A is a subset $C \subset A \times A$, written xCy. An **equivalence relation** $\sim$ is reflexive, symmetric and transitive; the **equivalence class** of x is $E = \{y : y \sim x\}$.

Lemma 1.2 (3.1)

Two equivalence classes E, E' are either disjoint or equal.

Proof. If $y \in E \cap E'$ with E the class of x and E' the class of x', then $y \sim x$ and $y \sim x'$, so $x \sim x'$. Any $w \sim x$ then satisfies $w \sim x'$; so $E \subset E'$ and symmetrically. ∎

Hence the classes form a **partition** of A (disjoint nonempty sets with union A); conversely every partition comes from the relation "lying in the same piece".

An **order relation** (simple or linear order) $<$ satisfies: comparability ($x \neq y \Rightarrow x < y$ or $y < x$), nonreflexivity (no $x < x$) and transitivity. In an ordered set we have open intervals (a, b), immediate successors and predecessors, and the **dictionary order** on $A \times B$: $a_1 \times b_1 < a_2 \times b_2$ iff $a_1 < a_2$, or $a_1 = a_2$ and $b_1 < b_2$. Ordered sets A, B have the same **order type** if there is an order-preserving bijection. For $A_0 \subset A$, an **upper bound** b satisfies $x \leq b$ for all $x \in A_0$; the least one is the **supremum** $\sup A_0$, and dually $\inf A_0$. A has the **least upper bound property** if every nonempty subset bounded above has a supremum; this is equivalent to the greatest lower bound property.

1.4 The integers and the real numbers

Motivation. We record the axioms of $\mathbb{R}$, chiefly the least upper bound property, and induction on $\mathbb{Z}_+$, so later proofs can cite exactly what they use.

We assume $\mathbb{R}$ is an ordered field with the least upper bound property; $\mathbb{Z}_+$ is the smallest inductive subset (containing 1 and closed under $x \mapsto x + 1$), which gives the principle of induction.

Theorem 1.3 (4.1, well-ordering property)

Every nonempty subset of $\mathbb{Z}_+$ has a smallest element.

Proof. First, for each n, every nonempty subset of $\{1, \ldots, n\}$ has a smallest element: the set A of n for which this holds contains 1, and if $n \in A$ and $C \subset \{1, \ldots, n+1\}$, then either $C = \{n+1\}$ or $C \cap \{1, \ldots, n\}$ has a least element, which is the least element of C. For a nonempty $D \subset \mathbb{Z}_+$ choose $n \in D$; the least element of $D \cap \{1, \ldots, n\}$ is the least element of D. ■

Theorem 1.4 (4.2, strong induction)

Let $A \subset \mathbb{Z}_+$ be such that $\{1, \ldots, n-1\} \subset A$ implies $n \in A$, for each n. Then $A = \mathbb{Z}_+$.

Proof. Otherwise let n be the least element of $\mathbb{Z}_+ - A$; then $\{1, \ldots, n-1\} \subset A$, so $n \in A$, a contradiction. ■

The **Archimedean ordering property** ($\mathbb{Z}_+$ is unbounded above in $\mathbb{R}$) follows from the least upper bound property: if $b = \sup \mathbb{Z}_+$ existed, $b - 1 < n$ for some n, so $n + 1 > b$.

1.5 Cartesian products

For an indexed family $\{A_\alpha\}_{\alpha \in J}$, the **cartesian product** $\prod_{\alpha \in J} A_\alpha$ is the set of all functions $x : J \to \bigcup A_\alpha$ with $x(\alpha) \in A_\alpha$ (written $(x_\alpha)_{\alpha \in J}$). For finite $J = \{1, \ldots, m\}$ these are m-tuples; for $J = \mathbb{Z}_+$ they are sequences; X^J denotes the set of all functions $J \to X$, e.g. $\mathbb{R}^\omega$, the set of real sequences.

1.6 Finite sets

Motivation. Finiteness is defined through bijections with $\{1, \ldots, n\}$, and its intuitive properties, including the pigeonhole principle, are proved.

A set is **finite** if it is empty or in bijection with a section $\{1, \ldots, n\}$; we show n is unique.

Lemma 1.5 (6.1)

Let $a_0 \in A$. There is a bijection $A \to \{1, \dots, n+1\}$ iff there is a bijection $A - \{a_0\} \to \{1, \dots, n\}$.

Proof. Given $g : A - \{a_0\} \to \{1, \dots, n\}$, extend by $a_0 \mapsto n+1$. Conversely, given $f : A \to \{1, \dots, n+1\}$, if $f(a_0) = n+1$ restrict; otherwise compose f with the transposition of $f(a_0)$ and $n+1$ first. ■

Theorem 1.6 (6.2)

Let $f : A \to \{1, \dots, n\}$ be a bijection and B a proper subset of A. Then there is no bijection $B \to \{1, \dots, n\}$, but (if $B \neq \emptyset$) there is a bijection $B \to \{1, \dots, m\}$ for some $m < n$.

Proof. Induction on n. For $n = 1$, A is a point and $B = \emptyset$. Assume the result for n and let $f : A \to \{1, \dots, n+1\}$, $B \subsetneq A$ nonempty. Pick $a_0 \in B$; $A - \{a_0\}$ is in bijection with $\{1, \dots, n\}$ by Lemma 1.5. Since $B - \{a_0\} \subsetneq A - \{a_0\}$, the induction hypothesis says $B - \{a_0\}$ is empty or in bijection with $\{1, \dots, p\}$, $p < n$, and not with $\{1, \dots, n\}$; by Lemma 1.5 again, B is in bijection with $\{1, \dots, p+1\}$, $p+1 < n+1$, and not with $\{1, \dots, n+1\}$. ■

Corollary 1.7 (6.3–6.5)

A finite set has no bijection with a proper subset of itself; $\mathbb{Z}_+$ is not finite; the cardinality of a finite set is uniquely determined.

Proof. The first is Theorem 1.6. $n \mapsto n+1$ is a bijection of $\mathbb{Z}_+$ onto a proper subset. If A were in bijection with $\{1, \dots, m\}$ and $\{1, \dots, n\}$, $m < n$, then $\{1, \dots, n\}$ would be in bijection with its proper subset $\{1, \dots, m\}$. ■

Corollary 1.8 (6.6)

A subset B of a finite set A is finite; if B is proper, its cardinality is smaller.

Corollary 1.9 (6.7)

For nonempty B the following are equivalent:
(1) B is finite;
(2) there is a surjection from some section $\{1, \dots, n\}$ onto B;
(3) there is an injection of B into some section.

Proof. (1)⇒(2) clear. (2)⇒(3): send b to the least element of $f^{-1}(b)$ (Theorem 1.3). (3)⇒(1): B is in bijection with a subset of a finite set; use Corollary 1.8. ■

Corollary 1.10 (6.8)

Finite unions and finite cartesian products of finite sets are finite.

Proof. For $A \cup B$ with $A \cong \{1, \dots, m\}$ and $B \cong \{1, \dots, n\}$, the map onto $\{1, \dots, m+n\}$ sending B to the last n values gives a surjection from a section; apply Corollary 1.9. $A \times B$ is the union of the finite sets $A \times \{b\}$; induct for more factors. ■

1.7 Countable and uncountable sets

Motivation. Countability criteria are needed for second countability and separability, and Cantor's diagonal argument shows not every infinite set is countable.

A set is **countably infinite** if it is in bijection with $\mathbb{Z}_+$, **countable** if finite or countably infinite, and **uncountable** otherwise.

Theorem 1.11 (7.1)

For nonempty B the following are equivalent:
(1) B is countable;
(2) there is a surjection $f : \mathbb{Z}_+ \to B$;
(3) there is an injection $g : B \to \mathbb{Z}_+$.

Proof. (1)⇒(2): if B is finite, extend a surjection from a section by a constant. (2)⇒(3): $g(b) = \min f^{-1}(b)$. (3)⇒(1): $g(B) \subset \mathbb{Z}_+$ is countable by the next lemma. ■

Lemma 1.12 (7.2)

An infinite subset $C \subset \mathbb{Z}_+$ is countably infinite.

Proof. Define $h(1) = \min C$ and $h(n) = \min\{C - h(\{1, \dots, n-1\})\}$, which is nonempty because C is infinite (recursive definition, Theorem 1.21). h is injective (increasing). It is surjective: if $c \in C$ were not in the image, then $h(n) < c$ for all n (c is always available), so h would map $\mathbb{Z}_+$ injectively into $\{1, \dots, c\}$, contradicting Corollary 1.7. ■

Corollary 1.13 (7.3)

A subset of a countable set is countable.

Corollary 1.14 (7.4)

$\mathbb{Z}_+ \times \mathbb{Z}_+$ is countably infinite.

Proof. $(n, m) \mapsto 2^n 3^m$ is injective by unique factorization; apply Theorem 1.11. ■

Theorem 1.15 (7.5)

A countable union of countable sets is countable.

Proof. Let $\{A_n\}_{n \in J}$, $J \subset \mathbb{Z}_+$, be countable and nonempty. Choose surjections $f_n : \mathbb{Z}_+ \to A_n$ (and, if J is finite, a surjection $g : \mathbb{Z}_+ \to J$). Then $(k, m) \mapsto f_{g(k)}(m)$ maps $\mathbb{Z}_+ \times \mathbb{Z}_+$ onto $\bigcup A_n$; compose with a surjection $\mathbb{Z}_+ \to \mathbb{Z}_+ \times \mathbb{Z}_+$. (The simultaneous choice of the f_n uses the axiom of choice.) ■

Theorem 1.16 (7.6)

A finite product of countable sets is countable.

Proof. $A \times B$ is the image of $\mathbb{Z}_+ \times \mathbb{Z}_+$ under $(n, m) \mapsto (f(n), g(m))$; induct on the number of factors. ■

Theorem 1.17 (7.7)

$\{0,1\}^\omega$ is uncountable.

Proof. Given $g : \mathbb{Z}_+ \to \{0,1\}^\omega$, write $g(n) = (x_{n1}, x_{n2}, \dots)$ and let $y_n = 1 - x_{nn}$. Then $y = (y_n)$ differs from $g(n)$ in the n-th coordinate, so g is not surjective. ■

Theorem 1.18 (7.8)

For any set A there is no injection $\mathcal{P}(A) \to A$ and no surjection $A \to \mathcal{P}(A)$.

Proof. An injection $\mathcal{P}(A) \to A$ would yield a surjection $A \to \mathcal{P}(A)$. Given $g : A \to \mathcal{P}(A)$, let $B = \{a : a \notin g(a)\}$. If $B = g(a_0)$, then $a_0 \in B \iff a_0 \notin g(a_0) = B$, a contradiction. ■

1.8 The principle of recursive definition

Motivation. The recursion principle justifies step-by-step definitions "choose x_1, then x_2, ...".

Fix a set C and the requirement $(*)$: $h(1) =$ the least element of C, and $h(i) =$ the least element of $C - h(\{1, \dots, i-1\})$ (with $C \subset \mathbb{Z}_+$ infinite, as in Lemma 1.12); the general case is identical.

Lemma 1.19 (8.1)

For each n there is a function $f : \{1, \dots, n\} \to C$ satisfying $(*)$ on its domain.

Proof. Induction: extend a function on $\{1, \dots, n\}$ by the value prescribed by $(*)$ at $n+1$ (the set $C - f(\{1, \dots, n\})$ is nonempty since C is infinite). ■

Lemma 1.20 (8.2)

Two functions on $\{1, \dots, n\}$ and $\{1, \dots, m\}$ satisfying $(*)$ agree on their common domain.

Proof. If not, let i be the least point of disagreement; $(*)$ determines the value at i from the (equal) values below i. ■

Theorem 1.21 (8.3–8.4, principle of recursive definition)

Let A be a set, $a_0 \in A$, and ρ a function assigning to each function f from a nonempty section $\{1, \dots, n\}$ into A an element $\rho(f) \in A$. Then there is a unique $h : \mathbb{Z}_+ \to A$ with $h(1) = a_0$ and $h(i) = \rho(h|_{\{1,\dots,i-1\}})$ for $i > 1$.

Proof. As in Lemmas 1.19 and 1.20, for each n there is a unique function on $\{1, \dots, n\}$ satisfying the formula, and these are compatible; their union is a function h on $\mathbb{Z}_+$ with the property, unique by the same argument. ■

1.9 Infinite sets and the axiom of choice

Motivation. The axiom of choice legitimizes infinitely many arbitrary choices, for instance a countably infinite subset of every infinite set.

Theorem 1.22 (9.1)

For a set A the following are equivalent:
(1) there is an injection $\mathbb{Z}_+ \to A$;
(2) there is a bijection of A with a proper subset of itself;
(3) A is infinite.

Proof. (1)⇒(2): if $f : \mathbb{Z}_+ \to A$ is injective, define $g(f(n)) = f(n+1)$ and $g(x) = x$ off $f(\mathbb{Z}_+)$; g is a bijection onto $A - \{f(1)\}$.

(2)⇒(3) is Corollary 1.7. (3)⇒(1): let c be a choice function on the nonempty subsets of A (Lemma 1.23) and define recursively $f(1) = c(A)$, $f(n) = c(A - f(\{1, \dots, n-1\}))$; the sets are nonempty because A is infinite, and f is injective. ■

Definition (Axiom of choice)

Given a collection $\mathcal{A}$ of disjoint nonempty sets, there is a set C consisting of exactly one element from each member of $\mathcal{A}$.

Lemma 1.23 (9.2, choice function)

Given a collection $\mathcal{B}$ of nonempty sets (not necessarily disjoint), there is a function $c : \mathcal{B} \to \bigcup \mathcal{B}$ with $c(B) \in B$ for each B.

Proof. The sets $B' = \{B\} \times B$, $B \in \mathcal{B}$, are disjoint and nonempty. A set C meeting each B' in one point is a set of pairs (B, x) with exactly one pair for each B; it is the desired function. ■

1.10 Well-ordered sets

Motivation. Well-ordered sets allow induction beyond $\mathbb{Z}_+$, and the minimal uncountable well-ordered set S_Ω is a source of counterexamples.

An ordered set is **well-ordered** if every nonempty subset has a smallest element. Subsets of well-ordered sets are well-ordered, and so are dictionary-ordered finite products of well-ordered sets. For α in a well-ordered set X, the **section** is $S_\alpha = \{x : x < \alpha\}$.

Theorem 1.24 (10.1)

Every nonempty finite ordered set has the order type of a section $\{1, \dots, n\}$; hence it is well-ordered.

Proof. Induction on the cardinality: remove the largest element (which exists in a finite ordered set, by induction) and append it at $n+1$. ■

Theorem 1.25 (Well-ordering theorem, Zermelo)

Every set can be well-ordered. (Equivalent to the axiom of choice.)

We accept this; a proof is a transfinite version of Theorem 1.22 using a choice function. In particular there exist uncountable well-ordered sets.

Lemma 1.26 (10.2)

There is a well-ordered set A with a largest element Ω such that the section S_Ω is uncountable but every other section of A is countable.

Proof. Let B be an uncountable well-ordered set and $C = \{1, 2\} \times B$ in dictionary order; some section of C is uncountable (e.g. that of $2 \times b$). Let Ω be the least element of C whose section is uncountable, and $A = S_\Omega \cup \{\Omega\}$. ■

S_Ω is called the **minimal uncountable well-ordered set**, and $\overline{S}_\Omega = S_\Omega \cup \{\Omega\}$.

Theorem 1.27 (10.3)

Every countable subset A of S_Ω has an upper bound in S_Ω.

Proof. $\bigcup_{a \in A} S_a$ is a countable union of countable sets, hence countable, so it is not all of S_Ω; any x outside it is an upper bound for A. ■

1.11 The maximum principle

Motivation. Zorn's lemma, equivalent to the axiom of choice, proves existence of maximal objects and is used for the Tychonoff theorem.

A **strict partial order** $\prec$ is nonreflexive and transitive; a subset on which it is a simple order is a **chain**.

Theorem 1.28 (11.1, Hausdorff maximum principle)

In a set with a strict partial order there is a maximal chain (a chain not properly contained in any other chain).

Proof. Well-order the set A by $<$ and define by transfinite recursion $h(\alpha) = \mathit{true}$ iff $\{\beta < \alpha : h(\beta) = \mathit{true}\} \cup \{\alpha\}$ is a chain for $\prec$. The set B of α with $h(\alpha)$ true is a chain (any two of its elements lie in the chain tested at the later one). It is maximal: if $B \cup \{\alpha\}$ were a chain for some $\alpha \notin B$, then $\{\beta < \alpha : \beta \in B\} \cup \{\alpha\}$ would be a chain, forcing $\alpha \in B$. ■

Theorem 1.29 (Zorn's lemma)

If A is strictly partially ordered and every chain has an upper bound in A, then A has a maximal element.

Proof. Take a maximal chain B and an upper bound c of B. If $c \prec d$ for some d, then $B \cup \{d\}$ is a chain properly containing B (note $d \notin B$ since $d \succ c \succeq$ every element of B), contradicting maximality. ■

Example 1.1

Every vector space has a basis: linearly independent sets under proper inclusion satisfy Zorn's hypothesis (a union of a chain is independent), and a maximal independent set spans. Zorn's lemma, the well-ordering theorem, the Hausdorff maximum principle and the axiom of choice are equivalent.